

Summary: Visualizations In Power BI - II

SESSION OVERVIEW:

By the end of this session, students will be able to:

- Create hierarchies in Power BI.
- Use custom visuals in Power BI.
- Utilise various types of charts as per need: scatter, pie, donut, funnel, gauge, ribbon, waterfall, bubble chart and treemap.
- Understand table, matrix, various types of maps and how to plot geographical data on maps.
- Choose the right kind of chart based on requirement.

KEY TOPICS AND EXAMPLES:

Note: For this session, we will be using [this dataset](#).

1. Creating a hierarchy in Power BI

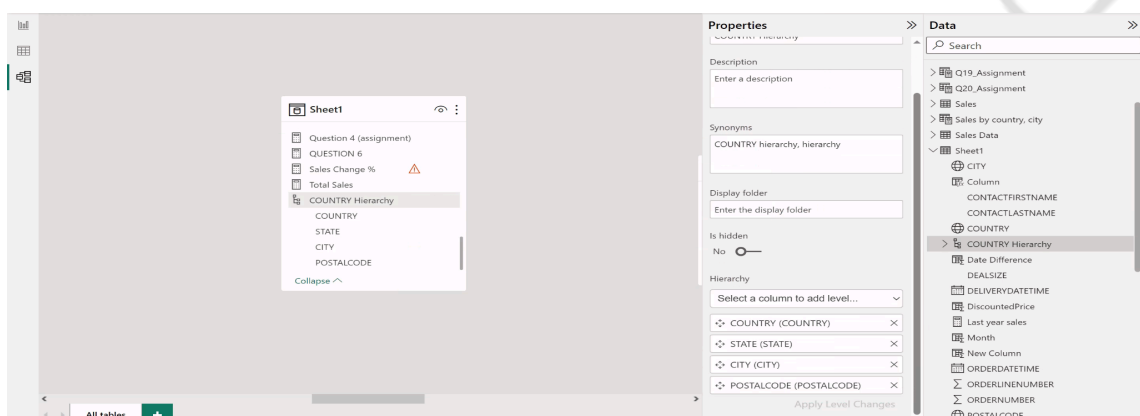
Power BI provides an option to create hierarchies for date and geographical data. A hierarchy aids in data visualisation and reporting. It is used to drill down from annual data to quarters, months, and days. As covered in a previous session, Power BI automatically creates a date hierarchy.

Steps to create a hierarchy:

- Go to the model view.
- In the table where you intend to create a hierarchy, right click on a column (e.g. Country in our dataset) and select Create Hierarchy.
- A new hierarchy gets created, which can be seen in the Properties pane on the right.
- In the properties pane, under the hierarchy section, select the other columns you want to add in your hierarchy (e.g. State, City, Postal Code in our dataset).
- Click on Apply Level Changes.

The new hierarchy gets created, as shown below. A geographic hierarchy like this one helps to **drill down** from Country, to State, City, and Postal Code.

Similarly, an organisational hierarchy can reflect the structure of the company from CEO to Director, Manager and Individual.



2. Visualizations in Power BI (contd...)

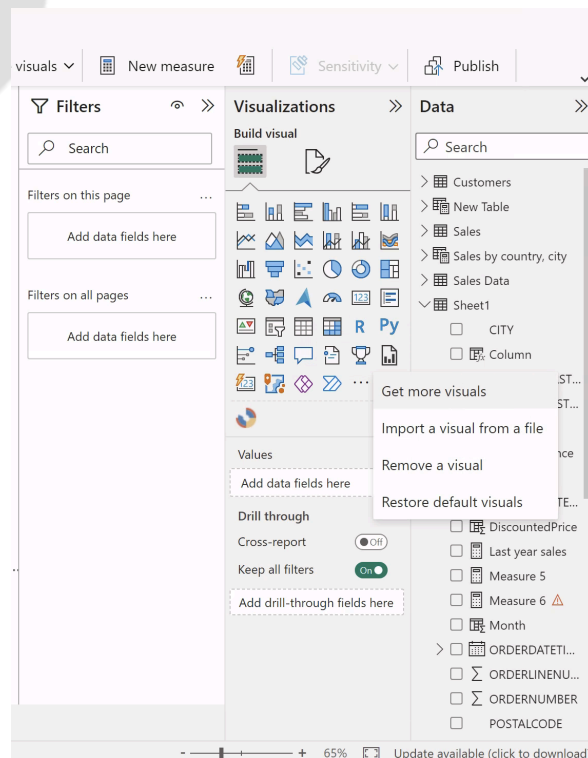
A. Custom Visuals

You can also get more visuals in Power BI, apart from the ones available in Power BI desktop. Examples include: Gantt Chart, Box and whisker chart, Time Series, Word Cloud, Animated bar graph and Sunburst.

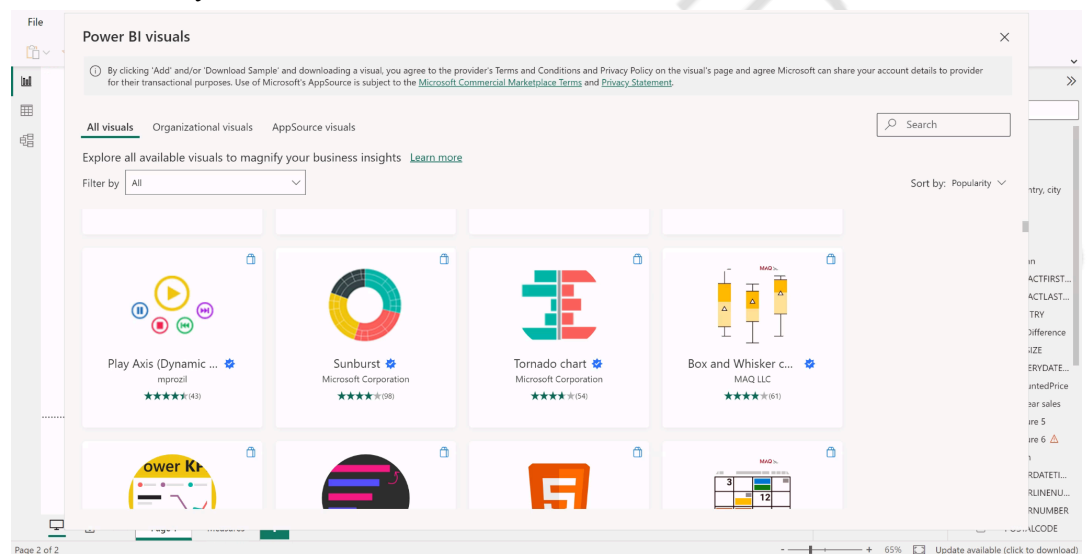
(Note: You need a Microsoft account to use custom visuals in Power BI.)

Steps required to use a custom visual are:

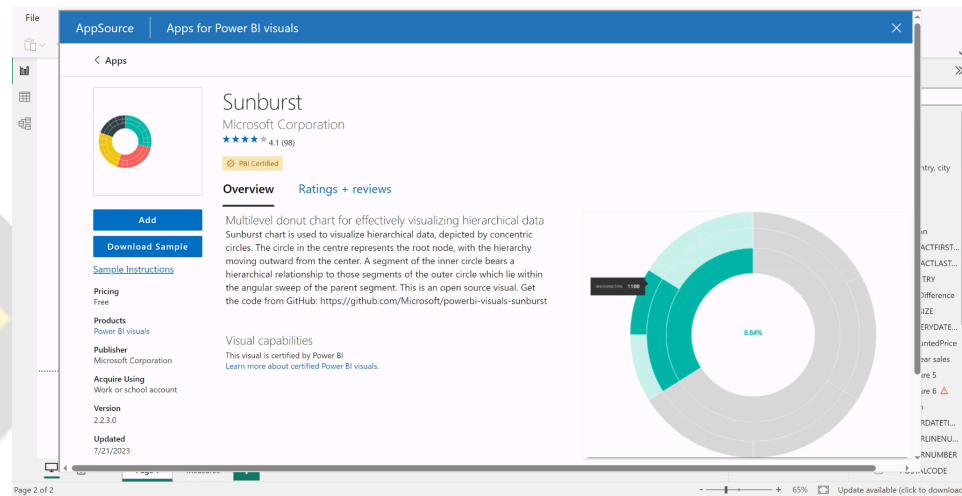
- In the visualizations pane, right click on the 3 dots and select Get more visuals.



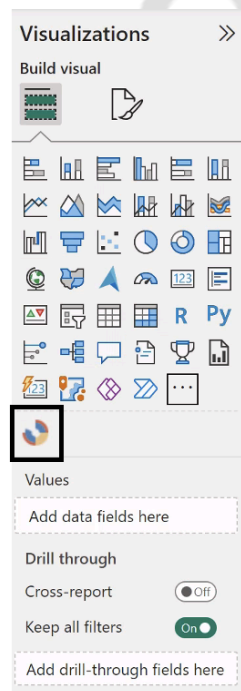
- Select the chart you want to add.



- Click on the 'Add' button and the chart will be added to the Visualizations pane.



- The visual will now appear in the visualization pane and can be used like a normal visual.



OR

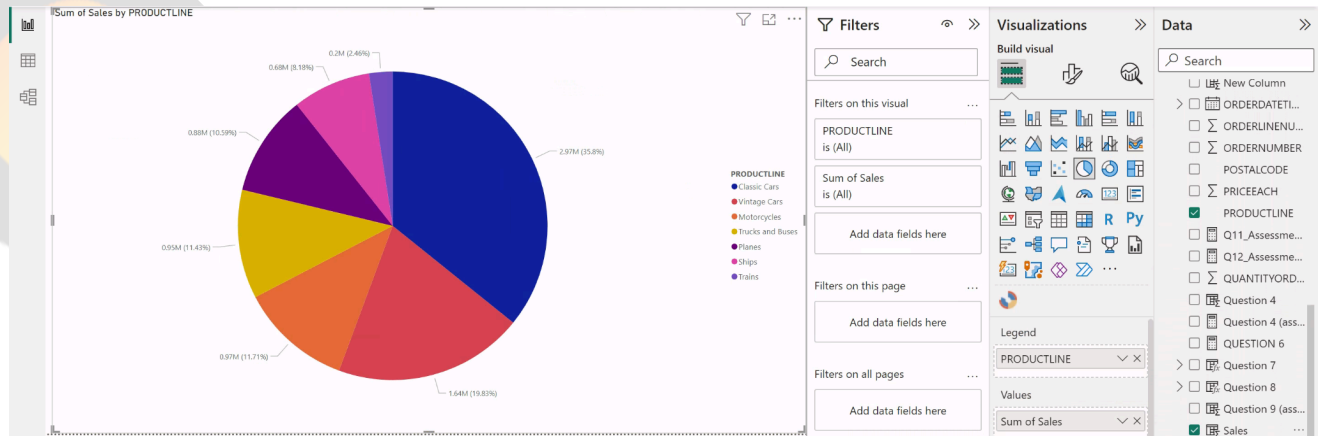
- Click Download a visual from community visuals in Power BI.
- In the Power BI visualization pane, click the 'Import Visual from a file' and upload the downloaded file.
- The visual will now appear in the visualization pane and can be used like a normal visual.

Note: We need to be cautious while using third party visuals due to possible security and performance issues.

B. Pie Chart

It shows the contribution of different values to a total. It tells us the percent of each pie out of the total. The sum of all values add up to 100%.

In our dataset, we can see the contribution of each PRODUCTLINE to the total sales using a pie chart.

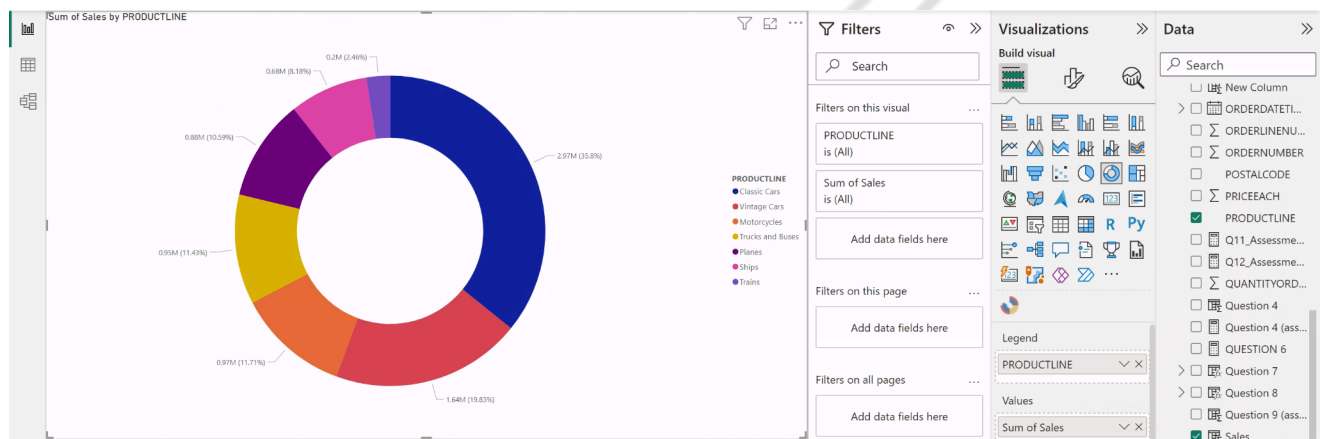


Here, we've kept the PRODUCTLINE in Legend and Sum of Sales in Values.

C. Doughnut Chart

They are similar to pie charts and show the contribution of different values to a total. The difference between a pie chart and a doughnut chart is that it is hollow in the center/the center is blank. Like a pie chart, the sum of all values add up to 100%.

In our dataset, we can see the contribution of each PRODUCTLINE to the total sales using a doughnut chart.

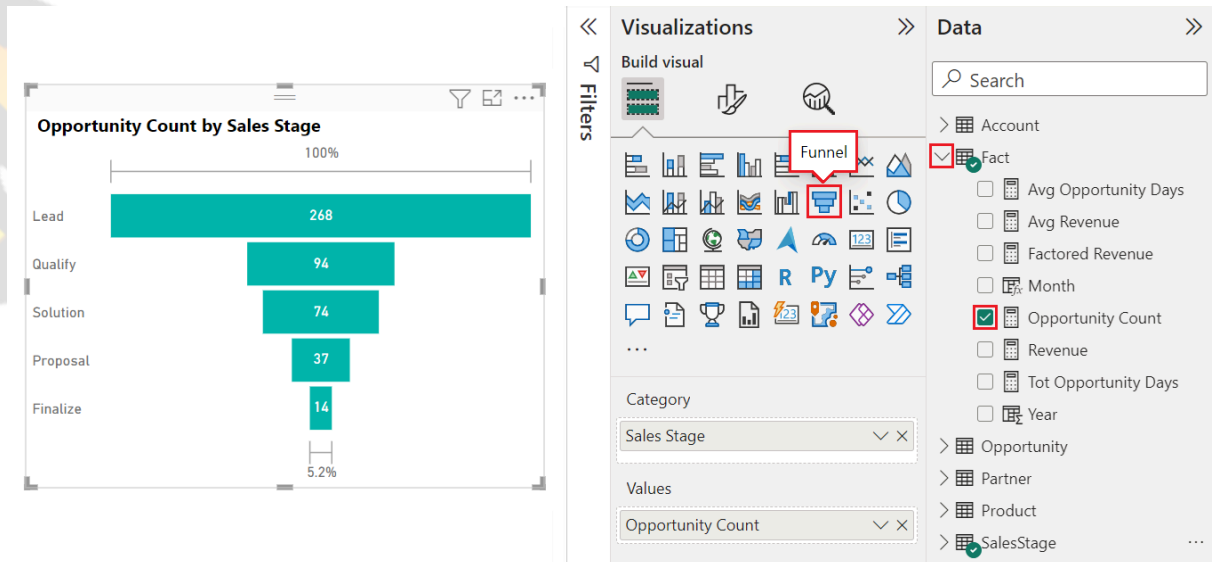


Here, we've kept the PRODUCTLINE in Legend and Sum of Sales in Values.

D. Funnel Chart

It is used to visualize a sequential process. It shows changes over a process such as sales conversion.

For example, in the figure given below - the typical steps involved in a sales funnel are:
Lead → Qualified Lead → Prospect → Contract → Close.

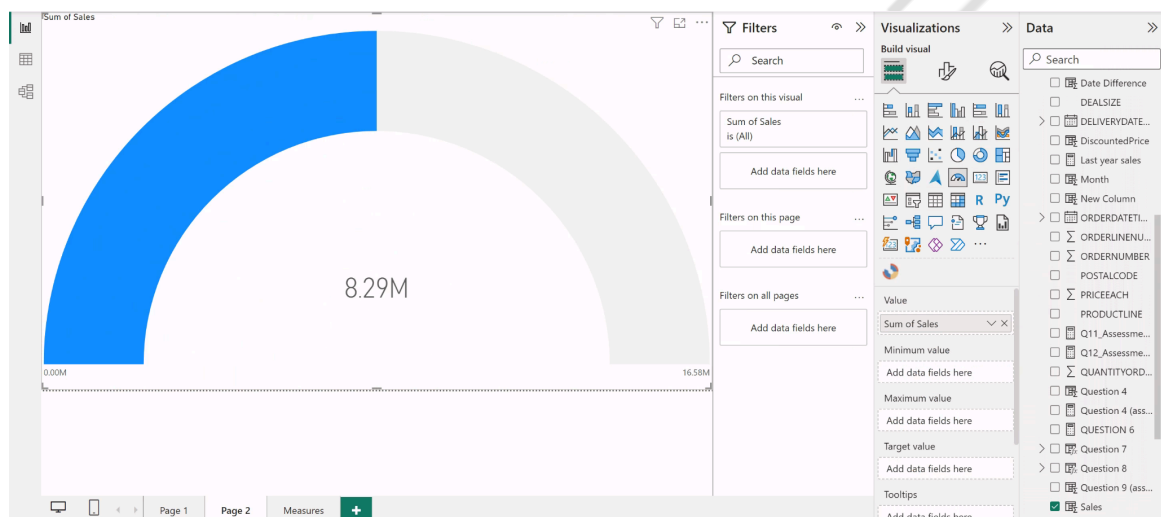


The above funnel chart shows that out of 268 leads, 14 were finally closed. In other words, 5.2% of the leads were finalised.

E. Gauge Chart

It is a semicircle chart that shows how a value progresses towards a goal/target value or Key Performance Indicator (KPI). The gauge line represents the goal/target value and the coloured region represents the progress towards the goal. In the arc, the value signifying the progress is enclosed.

By default (if we don't set a target value), the visual's maximum value will be the double of the actual value. We can set the minimum, maximum and target values in the 'Minimum value', 'Maximum value' and 'Target value' sections.



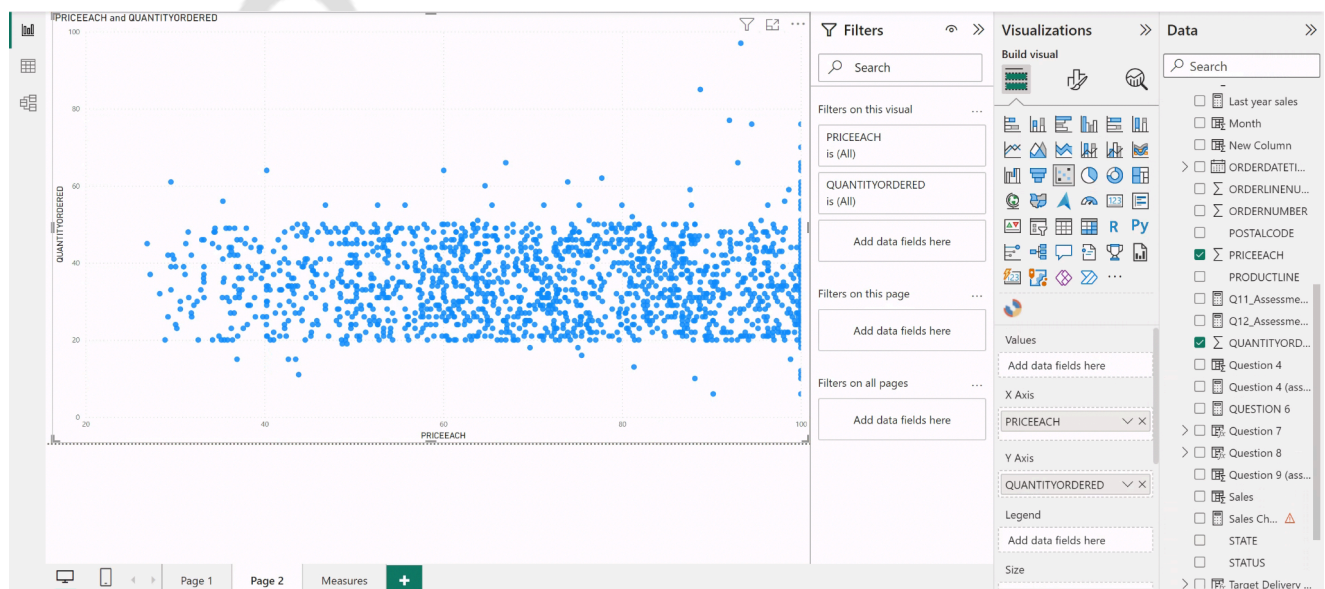
The chart above shows the Sales in our dataset. Because no target value is provided, it simply takes the double of the actual value.

F. Scatter Chart

It is used to compare 2 different measures on the same chart. It displays data along the horizontal axis (x-axis) and vertical axis (y-axis). It shows the relationship between two variables.

In our dataset, we can plot PRICEEACH and QUANTITYORDERED on the scatter plot to understand how price affects the quantity of order. For example, a lower price may lead to people ordering a higher quantity (although that does not seem to be the case with our dataset).

Here we have plotted PRICEEACH on x-axis and QUANTITYORDERED on y-axis.

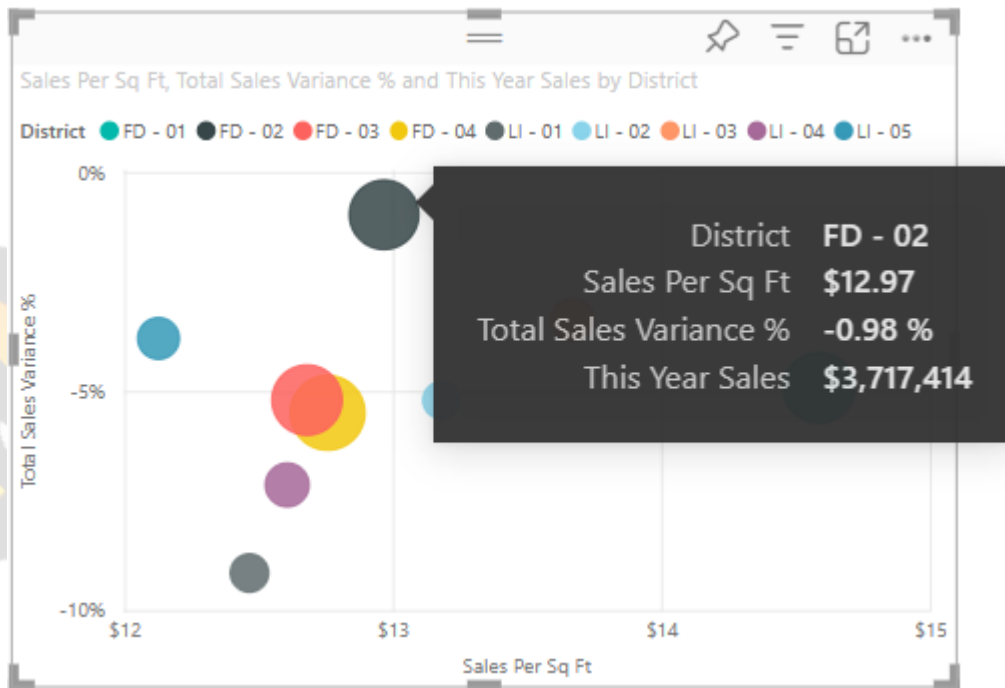


Another example of a situation where a scatter chart would be ideal is **Sales and Profit**. It answers the question: How does profit vary as sales increase?

G. Bubble chart

We can use the scatter plot to create a bubble chart as well. If we add a third dimension for Bubble size, a bubble chart will be created.

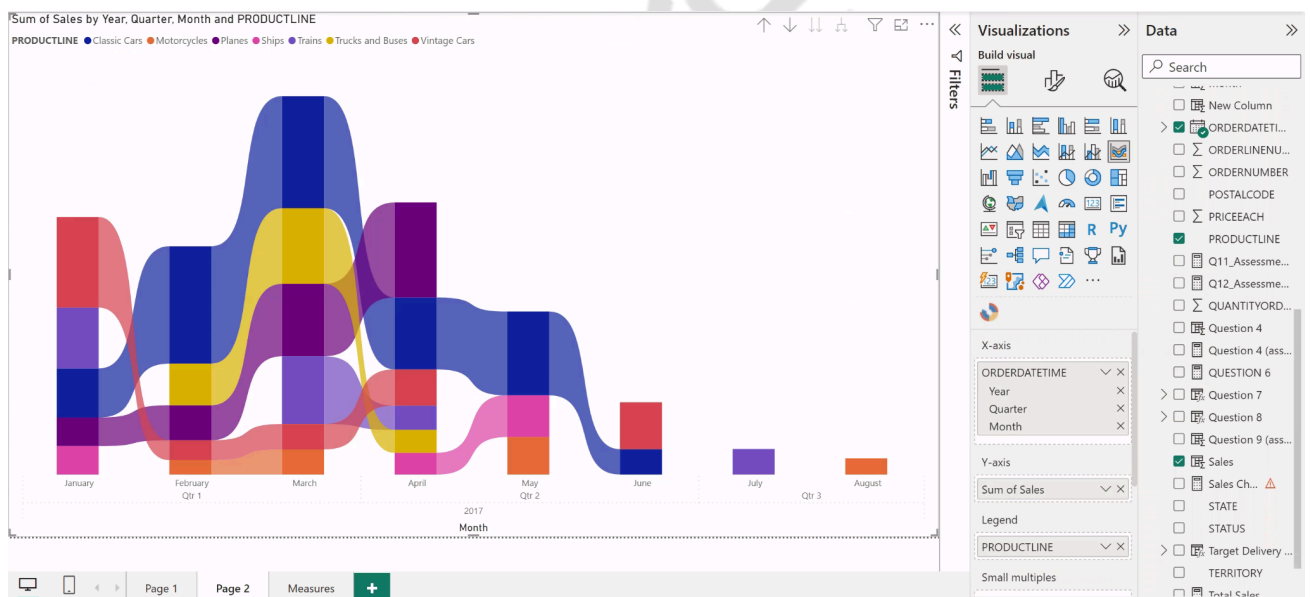
A bubble chart helps us visualize a third series of data using bubbles of different sizes. We can also display data in quadrants using a bubble chart.



H. Ribbon Chart

It shows which data category has the highest rank/largest value. Multiple categories are shown on the same chart in the form of ribbons of different colours.

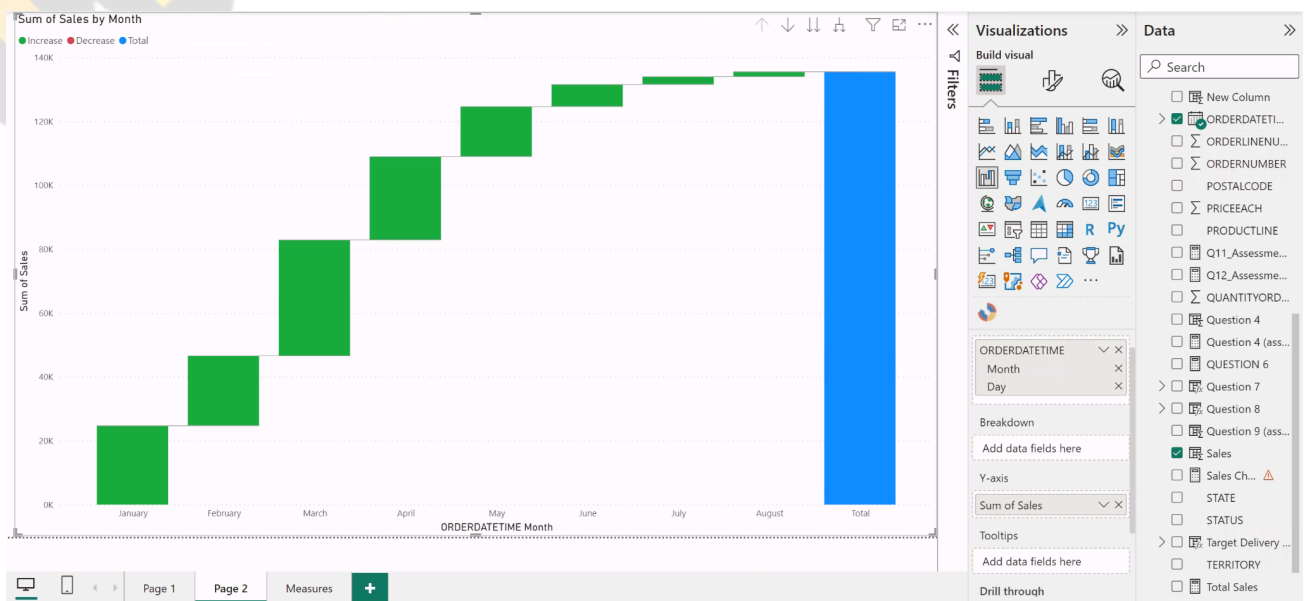
In our dataset, we can visualize total sales by PRODUCTLINE using the ribbon chart. We can use the PRODUCTLINE in the Legend field to create different ribbons for different product lines.



I. Waterfall Chart

It shows change in a particular value over time. It shows a running total as values are added and subtracted. It tells us how a value is affected by a series of positive and negative changes. Each change is a column on the chart.

In our dataset, we can see the change in sales every month using a waterfall chart. For the chart below we can see that the increase in sales in March was high as compared to other months while the increase in sales in July and August was very less.

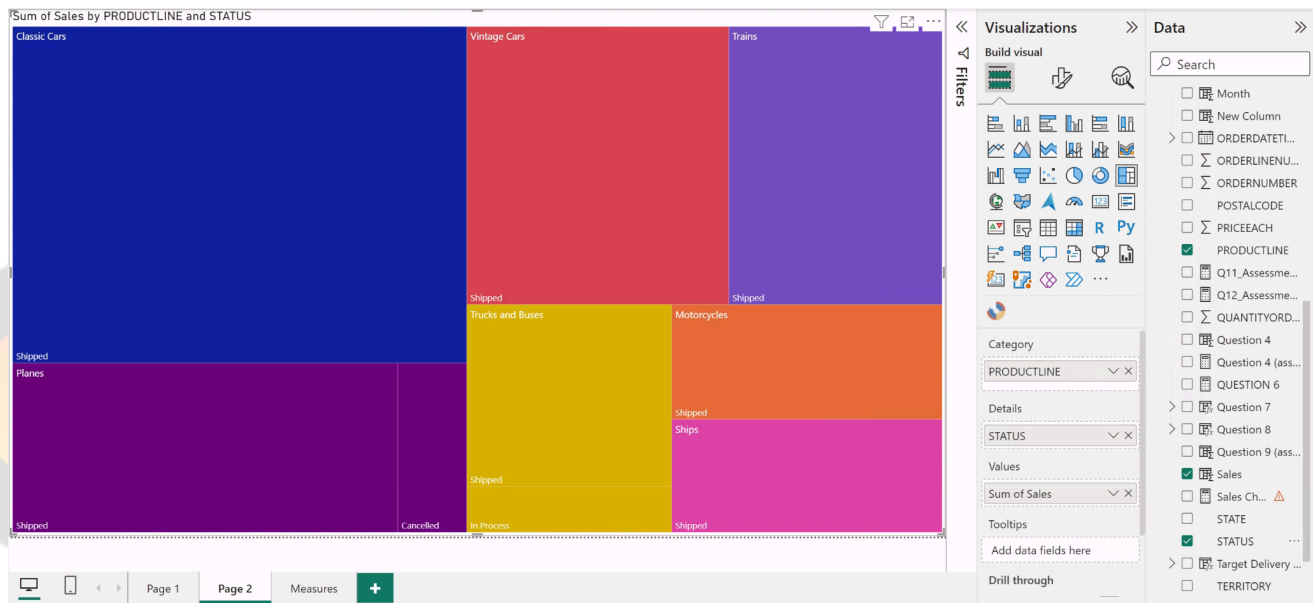


In a waterfall chart, a **category** is usually a **time-based field** and the y-axis is the value whose trend we intend to see.

J. Treemap

It is a hierarchical chart that shows the distribution of parent and child data. It displays data in the form of nested rectangles. Each level of the hierarchy is shown by a colored rectangle called a branch node. Each branch node contains smaller rectangles which are called leaf nodes. The value of a measure determines the size of each rectangle.

For example, in our dataset, the size of the rectangle (Values) is determined by the sum of sales. The category or the parent is the **PRODUCTLINE**, also called the branch node. The child data or the leaf node (Details) is the **STATUS** of the order.



K. Map

Note: To utilize map visualizations in Power BI, we need to first change the data type in Power BI. We need to assign a geographical data type to the data that we want Power BI to read as geographical data. This can be done using the following steps:

- Open Table View in Power BI.
- Select the column whose data type needs to be changed.
- Open the Column Tools tab in the ribbon.
- Select the Data Category option and select the desired data type from the dropdown list as shown below.

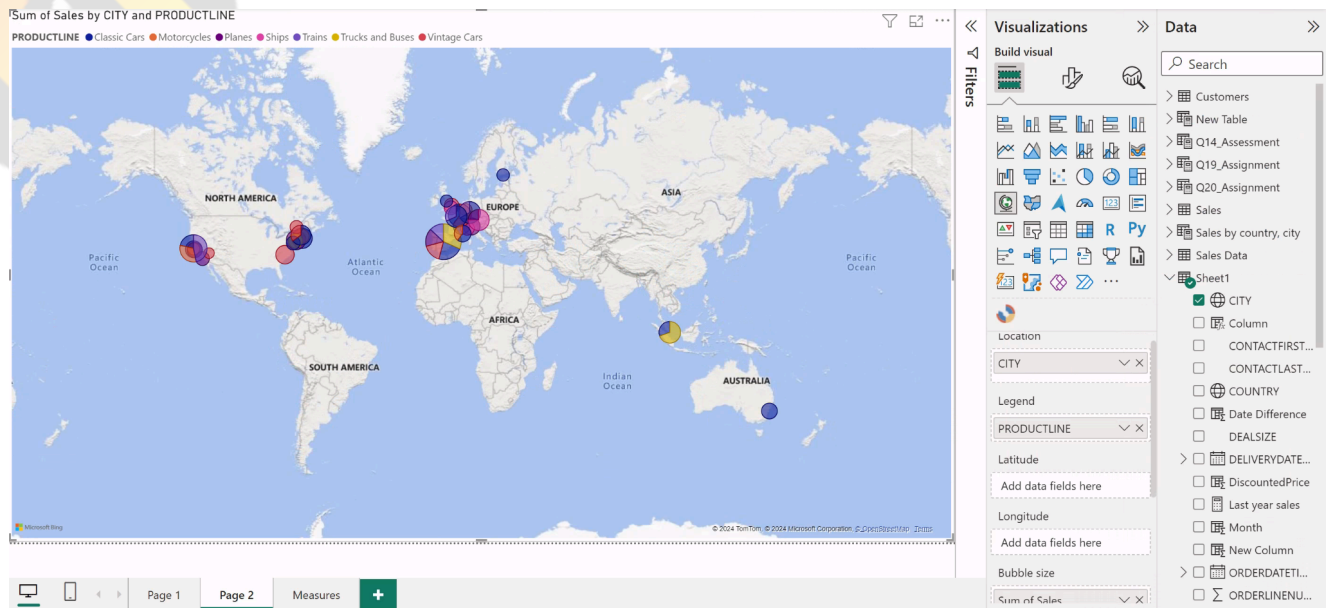
Once we define a geographical data category for the variable, a globe icon will appear next to the field name.

File Home Help External tools Table tools Column tools										
Name	CITY	Text								
ORDERNUMBER	QUANTITYORDERED	PRICEEACH	ORDERLINENUMBER	ORDERDATETIME	DELIVERYDATETIME	STATUS	PRODUCTLINE	CITY		
10405	97	93.28	5	11/12/2015 11:00:00 PM	11/15/2015 3:35:53 AM	Shipped	Classic Cars	Strasbourg		
10407	76	100	2	9/26/2012 1:01:00 AM	9/26/2012 11:42:17 PM	On Hold	Vintage Cars	San Jose		
10405	76	100	3	3/31/2012 12:50:00 AM	4/1/2012 2:20:11 AM	Shipped	Classic Cars	Strasbourg		
10401	85	88.75	10	5/11/2013 11:29:00 PM	5/12/2013 1:40:00 PM	On Hold	Planes	Newark		
10407	76	94.5	6	8/10/2013 12:03:00 AM	8/11/2013 3:11:39 AM	On Hold	Classic Cars	San Jose		
10401	77	92	9	5/30/2014 12:24:00 AM	5/31/2014 8:27:23 PM	On Hold	Planes	Newark		
10412	70	100	10	7/1/2016 1:45:00 PM	7/1/2016 10:11:13 PM	Shipped	Trucks and Buses	Madrid		
10419	70	100	8	8/9/2014 11:14:00 PM	8/11/2014 4:57:28 AM	Shipped	Classic Cars	Salzburg		
10403	66	100	9	12/13/2015 10:59:00 PM	12/14/2015 2:16:30 PM	Shipped	Motorcycles	Liverpool		
10403	66	100	6	3/20/2013 12:05:00 AM	3/21/2013 8:56:50 AM	Shipped	Motorcycles	Liverpool		
10417	66	100	2	6/3/2014 1:01:00 AM	6/5/2014 4:33:45 AM	Disputed	Motorcycles	Madrid		
10406	65	100	1	5/20/2016 12:36:00 AM	5/20/2016 5:11:55 PM	Disputed	Classic Cars	Kobenhavn		
10400	64	100	9	2/19/2012 2:16:00 AM	2/20/2012 3:18:20 AM	Shipped	Classic Cars	San Jose		
10420	66	92.95	6	6/23/2012 12:26:00 AM	6/25/2012 6:58:12 AM	In Process	Vintage Cars	Chatswood		
10406	61	100	3	7/31/2013 12:49:00 AM	8/2/2013 7:43:36 PM	Disputed	Classic Cars	Kobenhavn		
10412	60	100	9	8/28/2012 12:33:00 AM	8/30/2012 4:57:30 AM	Shipped	Classic Cars	Madrid		
10414	60	100	5	5/18/2016 10:31:00 PM	5/20/2016 7:09:26 AM	On Hold	Vintage Cars	Boston		
10407	59	100	11	2/14/2016 11:03:00 PM	2/16/2016 12:49:23 AM	On Hold	Classic Cars	San Jose		
10407	59	98.65	5	12/3/2012 12:34:00 AM	12/3/2012 5:55:44 PM	On Hold	Classic Cars	San Jose		
10400	58	100	6	6/17/2013 12:50:00 AM	6/19/2013 10:15:23 PM	Shipped	Vintage Cars	San Jose		
10417	56	100	4	6/9/2012 12:05:00 AM	6/10/2012 9:52:36 PM	Disputed	Motorcycles	Madrid		
10341	55	100	8	2/4/2016 1:22:00 AM	2/6/2016 5:34:52 AM	Shipped	Motorcycles	Salzburg		
10405	55	100	1	4/4/2016 1:12:00 AM	4/4/2016 11:18:45 AM	Shipped	Classic Cars	Strasbourg		
10419	55	100	2	4/23/2013 12:56:00 AM	4/25/2013 2:38:11 AM	Shipped	Classic Cars	Salzburg		
10340	55	100	2	1/16/2015 1:04:00 AM	1/16/2015 7:03:08 AM	Shipped	Motorcycles	Barcelona		
10342	55	100	7	6/15/2014 12:43:00 AM	6/16/2014 2:08:02 PM	Shipped	Classic Cars	Melbourne		
10339	55	100	13	9/29/2013 1:59:00 AM	9/30/2013 5:37:15 PM	Shipped	Vintage Cars	Minato-ku		

A map is used to visualize geographical data. It can be used to plot data as bubbles of different sizes on the map.

For example, from our dataset, if we want to see the sales in each city, we can use the following fields:

- Location: CITY
- Legend: PRODUCTLINE
- Bubble Size: Sum of Sales

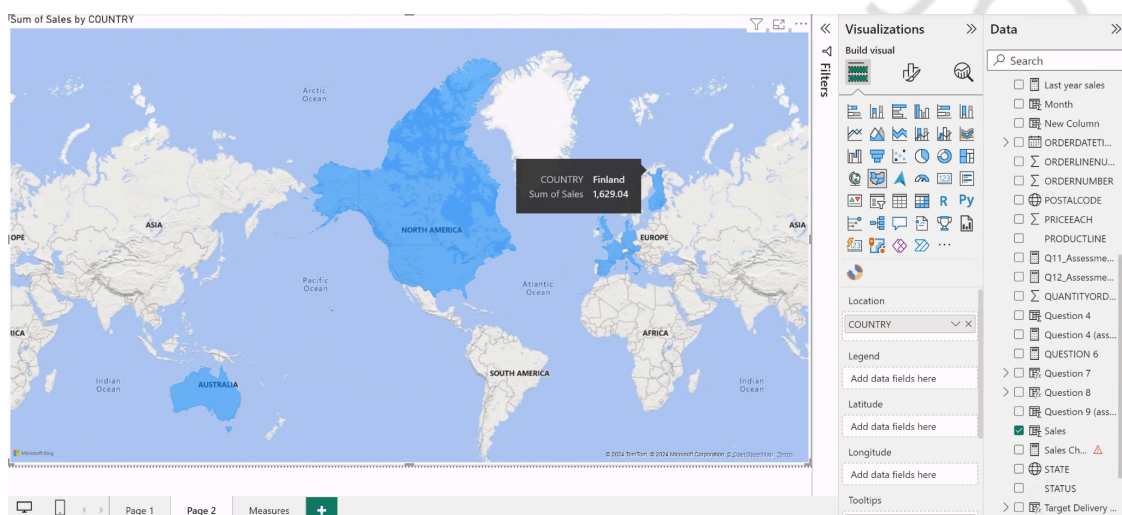


The size of the bubble in each city represents the total sales in that city and the pie chart in each bubble represents the contribution of different product lines to total sales.

L. Filled Map

It uses shading to display how a value differs across a geography. We can display shading differences from a light to dark colour.

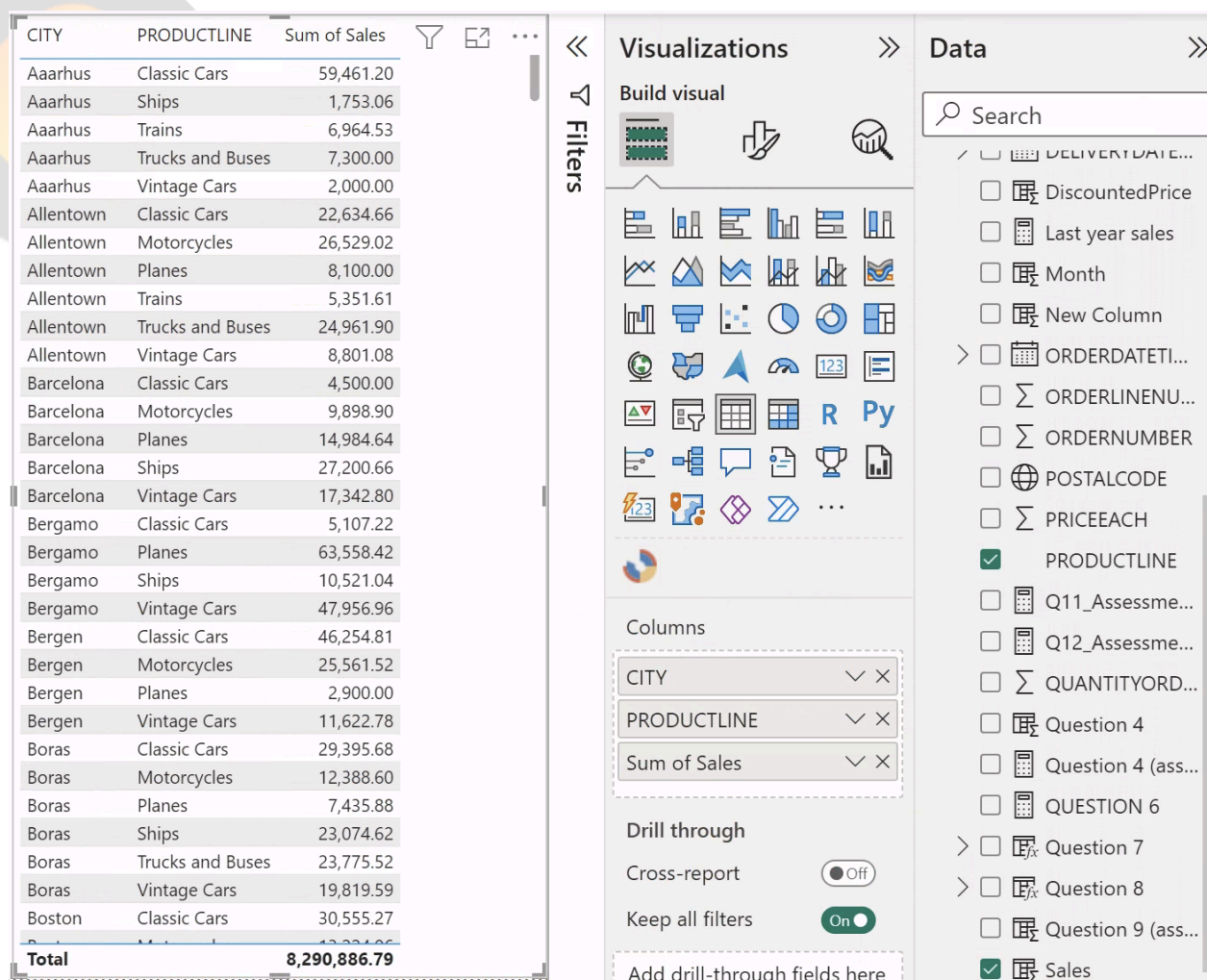
In our dataset, we can create a filled map which shows the sum of sales in each country.



M. Table

The table visual is very useful if we simply want to create a table with some fields. It can be used to aggregate/summarize values easily.

For example, if we want to see the total sales for each city and each product line, we can simply create a table.



The screenshot shows a data visualization tool interface. On the left is a table with columns CITY, PRODUCTLINE, and Sum of Sales. The table lists sales data for various cities and product lines, with a total sum of 8,290,886.79. In the center is the 'Visualizations' panel, which includes a 'Build visual' section with various chart icons, a 'Columns' section with a list of fields (CITY, PRODUCTLINE, Sum of Sales), and a 'Drill through' section with options for 'Cross-report' (Off) and 'Keep all filters' (On). On the right is the 'Data' panel, which contains a search bar and a list of fields including DELIVERDATE..., DiscountedPrice, Last year sales, Month, New Column, ORDERDATETI..., ORDERLINENU..., ORDERNUMBER, POSTALCODE, PRICEEACH, PRODUCTLINE (checked), Q11_Assessme..., Q12_Assessme..., QUANTITYORD..., Question 4, Question 4 (ass..., QUESTION 6, Question 7, Question 8, Question 9 (ass..., and Sales (checked).

CITY	PRODUCTLINE	Sum of Sales
Aarhus	Classic Cars	59,461.20
Aarhus	Ships	1,753.06
Aarhus	Trains	6,964.53
Aarhus	Trucks and Buses	7,300.00
Aarhus	Vintage Cars	2,000.00
Allentown	Classic Cars	22,634.66
Allentown	Motorcycles	26,529.02
Allentown	Planes	8,100.00
Allentown	Trains	5,351.61
Allentown	Trucks and Buses	24,961.90
Allentown	Vintage Cars	8,801.08
Barcelona	Classic Cars	4,500.00
Barcelona	Motorcycles	9,898.90
Barcelona	Planes	14,984.64
Barcelona	Ships	27,200.66
Barcelona	Vintage Cars	17,342.80
Bergamo	Classic Cars	5,107.22
Bergamo	Planes	63,558.42
Bergamo	Ships	10,521.04
Bergamo	Vintage Cars	47,956.96
Bergen	Classic Cars	46,254.81
Bergen	Motorcycles	25,561.52
Bergen	Planes	2,900.00
Bergen	Vintage Cars	11,622.78
Boras	Classic Cars	29,395.68
Boras	Motorcycles	12,388.60
Boras	Planes	7,435.88
Boras	Ships	23,074.62
Boras	Trucks and Buses	23,775.52
Boras	Vintage Cars	19,819.59
Boston	Classic Cars	30,555.27
Total		8,290,886.79

N. Matrix

A matrix is like a pivot table that creates a summarization of rows and columns based on the values provided.

In the example discussed above, Table visualization, can be more effectively visualized using a Matrix visual as shown below.

CITY	Classic Cars	Motorcycles	Planes	Ships	Trains	Trucks and Buses	Vintage Cars	Total
Aarhus	59,461.20			1,753.06	6,964.53	7,300.00	2,000.00	77,478.79
Allentown	22,634.66	26,529.02	8,100.00		5,351.61	24,961.90	8,801.08	96,378.27
Barcelona	4,500.00	9,898.90	14,984.64	27,200.66			17,342.80	73,927.00
Bergamo	5,107.22		63,558.42	10,521.04			47,956.96	127,143.64
Bergen	46,254.81	25,561.52	2,900.00				11,622.78	86,339.11
Boras	29,395.68	12,388.60	7,435.88	23,074.62		23,775.52	19,819.59	115,889.89
Boston	30,555.27	13,224.06		30,201.76	7,344.75	27,813.77	24,782.93	133,922.54
Brickhaven	61,115.25	15,709.27	17,718.98	12,826.08			33,775.96	141,145.54
Bridgewater	37,934.28	27,285.73	9,352.01		4,449.39			79,021.41
Brisbane	5,146.51					6,731.65	27,636.18	39,514.34
Bruxelles	3,000.00			26,490.72	6,555.41		32,522.05	68,568.18
Burbank	3,500.00		7,120.96			4,600.00	26,285.27	41,506.23
Burlingame	9,900.00	31,369.34	32,337.59			22,852.82	6,029.38	102,489.13
Cambridge	20,993.81	36,132.52	7,699.92	3,838.00		26,693.06	18,009.45	113,366.76
Charleroi	11,700.00	4,100.00	2,312.24	1,711.26			6,137.20	25,960.70
Chatswood	32,722.57	2,780.90	21,248.38	1,089.36		14,837.51	60,544.66	133,223.38
Cowes	3,300.00	22,626.74	25,346.55	4,598.40			17,298.44	73,170.13
Dublin	19,057.82	4,696.32	10,436.10		3,112.60	3,700.00	2,234.40	43,237.24
Espoo	58,747.72	12,584.73			5,117.05	12,303.76		88,753.26
Frankfurt	41,813.08				5,043.42	8,100.00	11,988.87	66,945.37
Gensve	93,344.91							93,344.91
Glen Waverly	10,714.30		3,843.84	3,070.40			38,511.08	56,139.62
Glendale	6,400.00	6,889.40	34,214.88	4,207.66			9,611.29	61,323.23
Graz	18,758.07					18,062.18	9,684.95	46,505.20
Helsinki	30,415.36	25,034.18	20,028.02	6,407.92			7,313.67	89,199.15
Kobenhavn	53,882.26		7,000.00	33,077.61	3,800.00		17,508.97	115,268.84
Koln	71,544.63	7,497.50						79,042.13
Las Vegas	53,297.67						19,531.38	72,829.05
Lille	6,391.92	50,464.47						56,856.39
Liverpool	53,502.42	32,055.91					7,692.26	93,250.59
London	21,647.35		13,949.18	23,920.09		25,280.40	25,112.35	109,909.37
Los Angeles	23,918.90		5,680.15	5,727.43			6,767.30	42,093.78
Lule	20,981.94			6,440.00	3,200.00	15,786.92	11,965.35	58,374.21
Total	2,968,546.40	971,086.29	877,942.21	677,940.40	203,804.26	947,355.18	1,644,212.05	8,290,886.79

It has fields just like pivot tables: Rows, Columns and Values.

2. When to use which type of chart?

It is important to identify which chart needs to be used for which kind of data. A few considerations for identifying the right chart are given below:

1. **Data type:** It is important to understand if the data is quantitative (numerical) or qualitative (categorical). Are we considering trends over time or simply comparing multiple categories.
2. **Goal:** We need to identify the goal of creating the visualisation. A very useful method here is to identify the question that needs to be answered.

Example 1: How did rainfall vary over time in the last month for 5 cities?

In this case, a simple line chart would do. There would be 5 lines and each line would represent one city. X-axis would have time.

Example 2: How did rainfall vary across 5 cities in the last month?

For this, we will create a bar/column chart with each bar/column representing one city.

Example 3: How did rainfall in each of 5 cities contribute to the total rainfall?

In this case, we will create a pie chart highlighting the % contribution of each city's rainfall to the total rainfall (sum of rainfall across all 5 cities).

Example 4: Display the comparative amount of rainfall across various cities in India along with their location.

The natural choice in this case is a map/filled map. The shade of the colour (dark shade/light shade) or the size of the bubble can represent the amount of rainfall.

3. **Conciseness:** In many situations we will find more than one chart that will be able to explain the data. We need to choose the one that is easiest to understand, clear and most concise.

Some types of charts and their best use:

1. **Area chart:** It is used for understanding the magnitude of change and total value over time.
2. **Line chart:** It is best for visualizing continuous data and trends over time.
3. **Bar/column chart:** It is best for visualizing and comparing a value (e.g. sales) across categories (e.g. product).
4. **Cards:** It is used for displaying values of metrics/measures/KPIs.
5. **Pie/Donut chart:** It is best for visualizing the contribution/% of each category.
6. **Scatter Plot:** It is useful for identifying the relationship between two variables. Also useful for identifying outliers and correlations.
7. **Gauge chart:** It is used to visualise how far the current value is from target value/goal.
8. **Map:** It is used to visualise data on a map when geographical data is there.